Invoice Provider API – Things to Know

Single API to create invoices

Creates an invoice for one or more freight documents

Async API – just because the API was successful doesn't mean it was sent to the shipper

API Endpoint: https://<BN4L API Hostname>/http/v1/LogisticsInvoice

Very important to send in the correct Sender System ID, Purchasing Party and Bill From Party

- Bill From Party and Purchasing Party are business partner Ids as defined in the shipper's TM system
- These values can be found in the Freight Order or Freight Booking
- These values are used to find the matching Freight Documents sent from the shipper

Note: If carrier submits invoice before shipper has sent the settlement – the invoice will be in status "Pending Submission" in the Manage Invoices app

Demo & Exercise

Invoice API

INTERNAL – SAP and Partners Only

Freight External Planning

Freight Collaboration

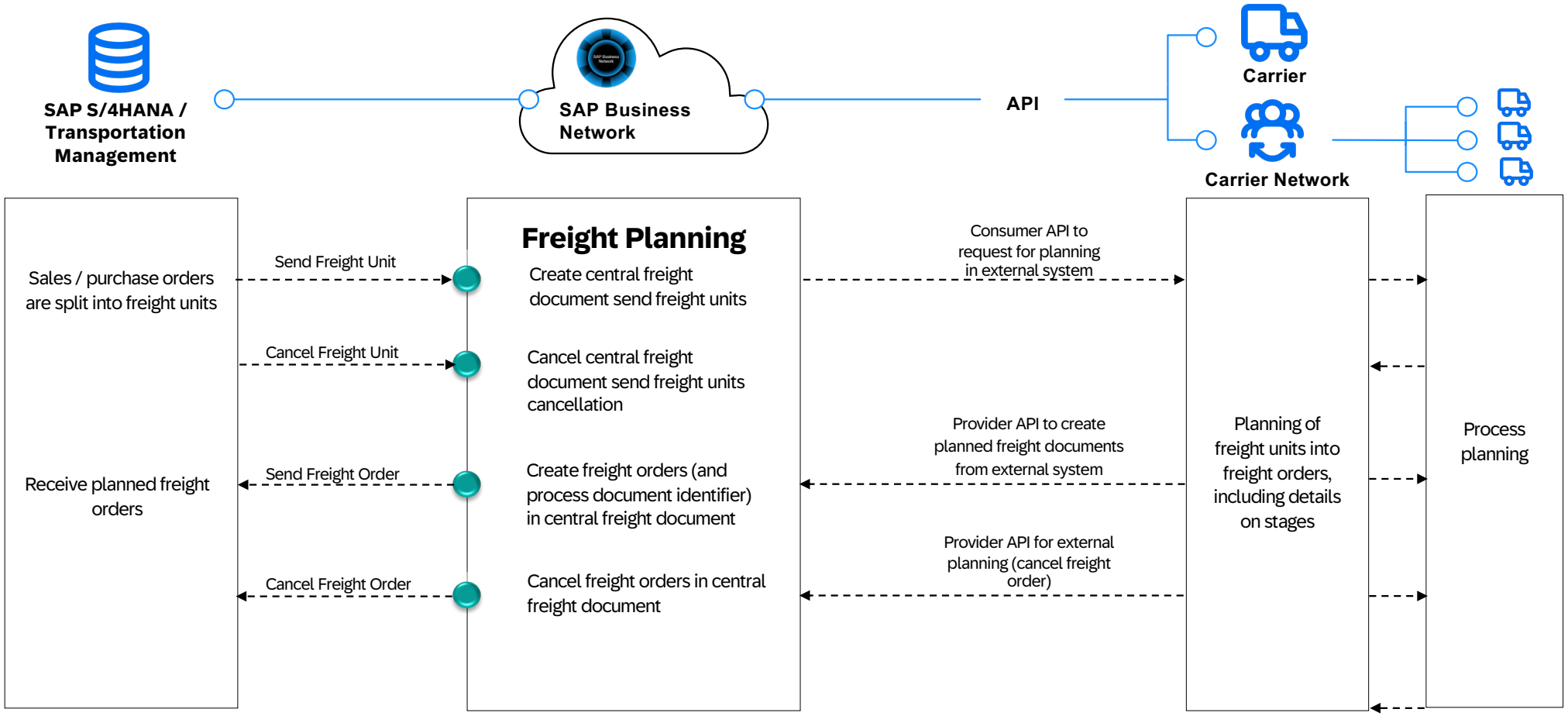
INTERNAL – SAP and Partners Only

Outsourced Planning help

External Planning blog

Leverage Freight Forwarders For External Planning

APIs to work directly from your SAP S/4HANA / Transportation Management System



Out-of-the-box integration with SOAP interface

Link to SAP API Hub for APIs

INTERNAL – SAP and Partners Only

DEMO



Agenda

Carrier Integration

- Integration Overview
- API Integration Details
- **EDI ANSI X12 Integration**
- EDI EDIFACT Integration
- Odds & Ends

Day 3

ANSI X12 EDI Overview

ANSI X12 EDI was in was invented in 1979 when bandwidth was expensive

- Very compact message format
- ~ signifies end of segment
- * is field delimiter
- Each space matters! Each field is given X amount of spaces
- First field in new row is the segment identifier. Ex: ISA
- Carriers who do EDI tend to know it very well

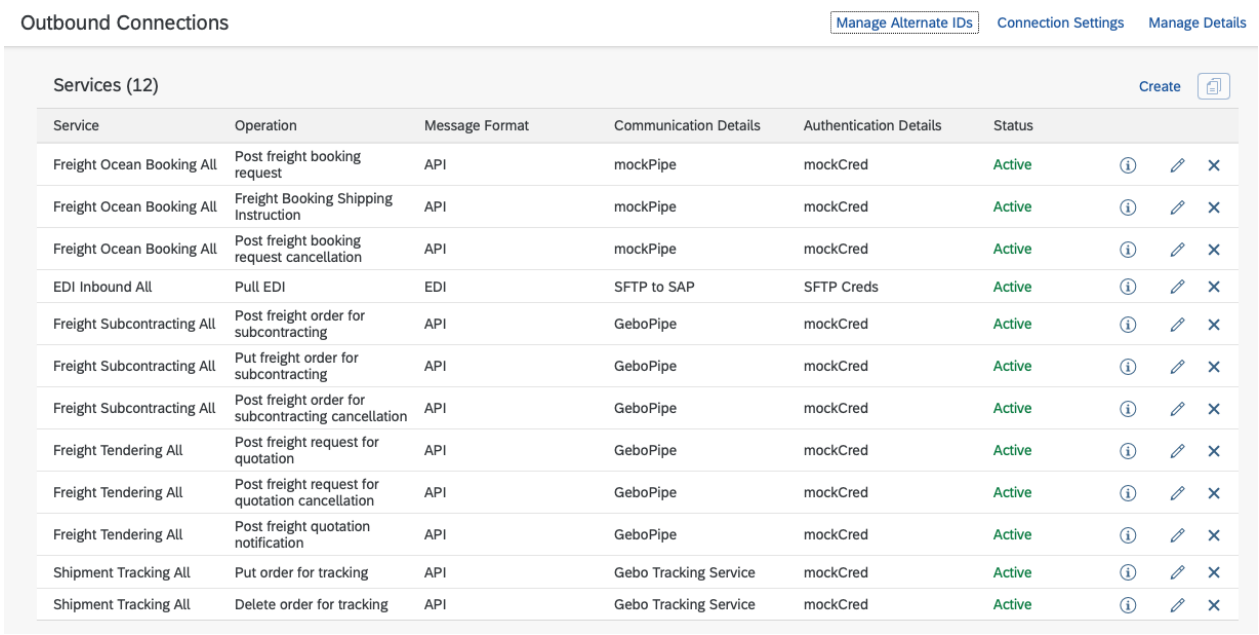
```
ISA*00*          *00*          *ZZ*10020005414    *ZZ*10020005430    *250331*1927*U*00401*000019862*0*P*>~
GS*SM*10020005414*10020005430*20250331*192730*000019862*X*004010~
ST*204*0001~
B2****6100002152**PP~
B2A*00*LT~
L11*S4HCLNT100*06~
L11*0017411710*CN~
L11*0017386001*12~
L11*5812*CO~
L11*80005746*DO~
NTE*TRA*513 KMT~
NTE*CBH*USD~
S5*1*CL~
L11*RES.TEST17186001*PO~
G62*69*20250407*U*0700*UT~
G62*69*20250407*U*0000*PT~
N1*SF*Shipping Point 1710 - - Address Name 1*93*SP_1710~
N3*3475 Deer Creek~
N4*Palo Alto*CA*94304-1355*US~
OID*6100002152*RES.TEST17186001~
L5*1*Trad.Good 11,PD,Reg.Trading~
AT8*G*K*190*10**X*2~
S5*2*CU~
G62*70*20250407*X*0700*UT~
G62*70*20250407*X*0000*PT~
N1*ST*US TM Customer11*93*0017186011~
N3*8989 Market Street~
N4*Long Beach*CA*90813*US~
OID*6100002152*RES.TEST17186001~
L5*1*Trad.Good 11,PD,Reg.Trading~
AT8*G*K*190*10**X*2~
L3*15190*G***106610****2*X**K~
SE*31*0001~
GE*1*000019862~
IEA*1*000019862~
¶
```

INTERNAL – SAP and Partners Only

System Connections – EDI Configuration

System Connections app is one stop shop for setting up carrier side integrations

- Main screen – define services you support (can also test on a test tenant)
- Connection Settings – define integration platform (always pick Multi-Cloud), and enable inbound connections
- Manage Details – define logical SFTP connections
- Manage Alternate IDs – see your tenants LBN ID, enter in alternate IDs like DUNS, SCAC, GLN, USDOT, etc
  – Not needed for carrier integrations

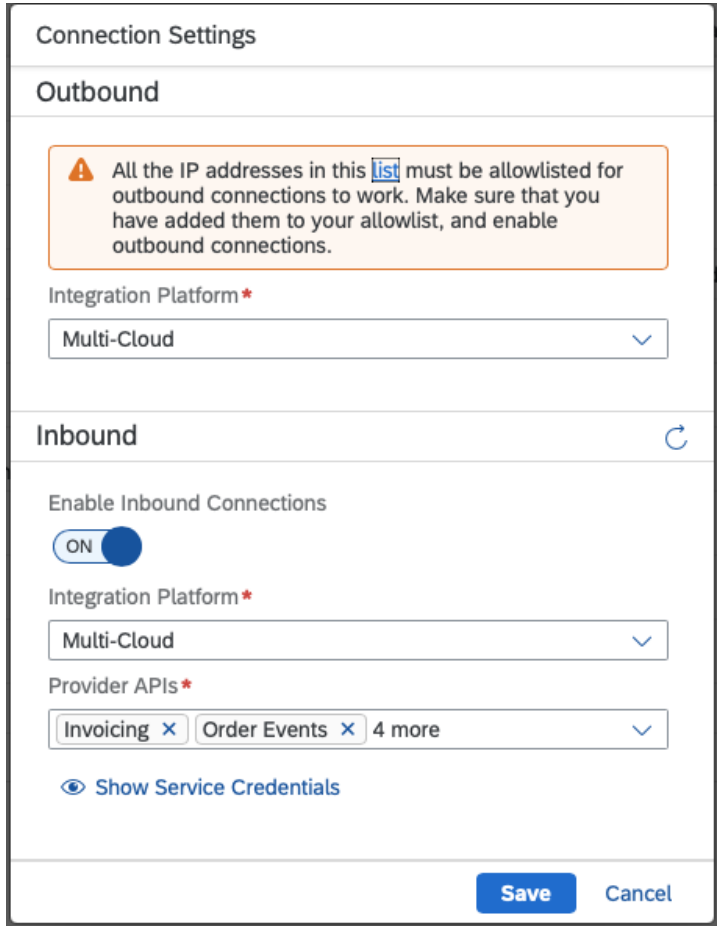


Service	Operation	Message Format	Communication Details	Authentication Details	Status
Freight Ocean Booking All	Post freight booking request	API	mockPipe	mockCred	Active
Freight Ocean Booking All	Freight Booking Shipping Instruction	API	mockPipe	mockCred	Active
Freight Ocean Booking All	Post freight booking request cancellation	API	mockPipe	mockCred	Active
EDI Inbound All	Pull EDI	EDI	SFTP to SAP	SFTP Creds	Active
Freight Subcontracting All	Post freight order for subcontracting	API	GeboPipe	mockCred	Active
Freight Subcontracting All	Put freight order for subcontracting	API	GeboPipe	mockCred	Active
Freight Subcontracting All	Post freight order for subcontracting cancellation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight request for quotation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight request for quotation cancellation	API	GeboPipe	mockCred	Active
Freight Tendering All	Post freight quotation notification	API	GeboPipe	mockCred	Active
Shipment Tracking All	Put order for tracking	API	Gebo Tracking Service	mockCred	Active
Shipment Tracking All	Delete order for tracking	API	Gebo Tracking Service	mockCred	Active

System Connections > Connection Settings – EDI Configuration

System Connections > Connection Settings where you configure integration platform and turn on inbound connectivity

- Integration Platform – **always pick Multi-Cloud** – Neo is deprecated. **Note**: if you know carriers still on Neo they need to migrate by mid May 2025!
- Inbound– turn on the switch if you want to send EDI responses
  – Select Multi-Cloud for integration platform
  – Select all provider APIs even though doing EDI



System Connections > Manage Details – API Configuration

System Connections > Manage Details where you define SFTP settings and Auth Details

- Communication Details – SFTP connection settings for outbound and inbound documents
  - Outbound: for each outbound document carrier plans to support. Example 204, 213
- Authentication Details – used to auth against SFTP server

Note: SFTP Polling Schedule for inbound documents (990, 214 & 210) must be requested through support portal

